Lab experiment

Travelling Salesman Problem

Coding:

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# Travelling Salesman Problem

from itertools import permutations

# Cost matrix
graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
cities = list(range(n))

# Find minimum cost tour
min_cost = float('inf')
best_path = []

for path in permutations(cities[1:]):
    tour = (0,) + path + (0,)

    cost = 0

    for i in range(len(tour) - 1):
        cost += graph[tour[i]][tour[i + 1]]

    if cost < min_cost:
        min_cost = cost
        best_path = tour

# Display result
print("Minimum Cost:", min_cost)

print("Best Tour:")
for city in best_path:
    print(city, end=" ")
```

Output:

```
Output

Minimum Cost: 80
Best Tour:
0 1 3 2 0
```